# Ashini Kavindya Nawagamuwa

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Ashini Kavindya | AshiniKavindya
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## Education

- **University of Moratuwa** — *Month Year - Mar 2024- Present*
  *B.Sc. Eng. (Honours) in Computer Science and Engineering (DSE Specialization)* — Moratuwa, Sri Lanka
  - GPA: 3.36/4.00
- **St.Joseph's Girls Collage** — *Jan 2023*
  *High School* — Kegalle, Sri Lanka
  - GCE Advanced Level-(2023) : Z-Score: +2.56, Island rank-179 out of 35,197
    - Subjects and Grades: Combined Mathematics (A), Chemistry (A), Physics (A)
  - GCE Ordinary Level(2019) : 8A 1B

## Projects

- **grapheme++** — *Jan 2026 - Present*
  *A library for more accuraate text processing for sinhala and tamil scripts*
  - Technologies: Python, Unicode text processing, grapheme segmentation, NLP metrics.
- **Medsync - Clinic Appointment and Treatment Management System** — *Nov 2025*
  *a full stack web application designed to streamline clinic operations, manage patient appointments, and handle treatment records efficiently.*
  - Designed secure authentication and authorization workflows to protect sensitive medical data.
  - Technologies: React (TypeScript), Node.js, Express.js, PostgreSQL, TailwindCSS
- **RPAL Interpreter** — *Mar 2026 – April 2026*
  *A Python-based interpreter for the RPAL programming language, implemented from scratch as part of coursework at University of Moratuwa.*
  - Built a custom lexical analyzer, recursive descent parser, and AST generator.
  - Applied transformation rules to standardize the AST and evaluated it via a Control Stack Environment (CSE) machine.
  - Matched outputs against the official rpal.exe reference and added CLI flags for debugging.
  - Technologies: c++,recursive descent parsing, tree-based evaluation.
  - GitHub: https://github.com/ashiniKavindya/RPAL-Interpreter
- **Sinhala Air Writing Recognition (1D-CNN)** — *Mar 2026 – Present*
  *A real time gesture recognition system translating spatial index-finger trajectories into Sinhala Unicode.*
  - Curated a custom time series gesture dataset for Sinhala characters and modifiers using a self developed video collection and tracking utility.
  - Built a spatial preprocessing pipeline using MediaPipe and SciPy to capture, zero center, and interpolate finger trajectories into uniform 100 point time series signals.
  - Designed a lightweight 1D-CNN in PyTorch optimized for low latency temporal feature extraction, enabling instantaneous real time character inference on edge devices.
  - Technologies: Python, PyTorch, MediaPipe, NumPy, SciPy, OpenCV
- **Sentiment Analysis System (Machine Learning / NLP Project)** — *Dec 2025*
  *Developed a sentiment analysis model to classify text data as positive or negative.*
  - Built an interactive web interface using HTML and JavaScript for user input and real time sentiment display.
  - Trained, tuned, and evaluated models on a Kaggle dataset using accuracy, precision, recall, and F1-score to ensure reliable performance.
  - Technologies: Python, Scikit-learn, NLP, Pandas, NumPy, HTML, JavaScript
- **Nano-Processor Simulation Project** — *April 2025*
  *Designed and implemented a 4-bit nanoprocessor using VHDL, with a custom instruction set, ALU, and control unit, verified through simulation and FPGA testing.*
  - Tools: Xilinx Vivado, VHDL, Assembly Language

## Publications

- **Trilingual Topic Modeling of Sri Lankan Parliamentary Debates** *2026*
  *Accepted at MERCon 2026 (Collaborative Research)*
  - Proposed an end-to-end multilingual topic modeling pipeline for Sri Lankan Hansard debates, integrating LLM based extraction, multilingual embeddings, dimensionality reduction, clustering, and topic representation.
  - Evaluated multilingual embedding models for long form code mixed parliamentary speeches using cross lingual semantic similarity, semantic retrieval, and anisotropy analysis.
  - Conducted a comparative study of clustering algorithms, demonstrating the effectiveness of HDBSCAN for high purity topic extraction while analyzing precision coverage trade-offs.

## Certifications

- **Advanced Learning Algorithms(coursera)** *Jan 2026*
- **Python for Data Science, AI and Development(coursera)** *Jan 2026*
- **Developing AI Applications with Python and Flask(coursera)** *Dec 2025*
- **Supervised Machine Learning: Regression and Classification(coursera)** *Dec 2025*
- **Data cleaning - Kaggle** *Feb 2026*
- **Pandas - Kaggle** *Jan 2026*
- **Intermediate Machine Learning - Kaggle** *Dec 2025*
- **intro to Machine Learning - Kaggle** *Dec 2025*

## Skills

- **Programming Languages:** Python, C++, Java, HTML, CSS, JavaScript
- **Web Technologies:** react, NodeJs

## References

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